

CoreAxisHR

HR Employee Control System

Phase 2 Progress Report | Oswaldo Cardona, Samuel Mendoza | Group 04 | April 27, 2026

1. Project Title & Group Members

Project: CoreAxisHR — HR Employee Control System
Course: Web Systems Development | Group 04 | Phase 2 Submission

Member	Role	Key Contributions
Alice Johnson	Team Lead / Full-Stack	Project architecture, login system, dashboard, session management
Bob Martinez	Back-End Developer	Employee CRUD (PHP + MySQL), leave request logic, database schema
Carol White	Front-End Developer	CSS design system, responsive layout, leave request UI
[Member 4]	Back-End / API	Brevo email API integration, leave_action.php, config management

Note: Replace [Member 4] with the actual name before submission.

2. Features Completed So Far

The following features have been fully implemented and are functional in the current build:

Feature	Status	Notes
User Login & Logout	Complete	Session-based auth; redirects on success or invalid credentials
Role-Based Access Control	Complete	Three roles (Admin, Manager, Employee) with conditional UI rendering
Employee Directory (CRUD)	Complete	Add, view, edit, delete employees; all operations hit MySQL
Leave Request Submission	Complete	Employees submit requests via form; stored in leave_requests table
Leave Approval / Rejection	Complete	Admins and Managers approve or reject pending requests
Email Notifications (Brevo)	Complete	HTML email sent on approval/rejection via Brevo REST API
JavaScript Live Search	Complete	Real-time client-side filter on employee directory table
JS Confirmation Dialogs	Complete	Confirm dialogs before all destructive/impactful actions
CoreAxisHR Branding / CSS	Complete	Consistent sidebar, topbar, card layout, and color palette
config.php Credential Safety	Complete	config.php excluded from repo; config.example.php provided

3. User Roles Implemented

Three distinct roles are stored in the users table and enforced at the PHP session level throughout every page:

Role	Demo Username	Permissions
Admin	alice	Full access — view and manage all employees, approve or reject any leave request, see all system data.
Manager	bob	Elevated access — view the employee directory and approve/reject leave requests; cannot add or delete employees.
Employee	carol	Restricted access — can only view and submit their own leave requests. The Employees menu item is hidden.

Role enforcement is applied on every protected page via a session check at the top of each PHP file. Unauthorized route access redirects to login.php or leave.php as appropriate.

4. Database & CRUD Status

The application uses a MySQL database named hr_test running on localhost via XAMPP. All queries are executed through the mysqli_* procedural API.

Table / Feature	Status	Notes
Table: employees	Complete	id, name, job_title, status — full Create/Read/Update/Delete
Table: users	Complete	id, username, password, role, employee_id — read-only in Phase 2
Table: leave_requests	Complete	id, employee_id, type, dates, reason, status, created_at — C/R/U
Foreign key: employee_id	Partial	Logical link present in queries; formal FK constraint not yet added
Password hashing	Pending	Passwords stored as plain text — to be replaced with password_hash()
Prepared statements	Pending	Raw string interpolation used; PDO/prepared statements planned for Phase 3

The database schema and sample data scripts (schema.sql and sample_data.sql) are included in the database/ folder, allowing any team member or instructor to reproduce the environment from scratch.

5. JavaScript Features Currently Working

All JavaScript is contained in js/app.js, loaded at the bottom of every page. Two core features are operational:

- Live Search on Employee Directory — A keyup event listener on the #searchInput field filters visible table rows in real time by comparing the search string against each row's full text content. No page reload is required.
- Confirmation Dialogs on Destructive Actions — Four functions (confirmDelete(), confirmAdd(), confirmApprove(), confirmReject()) each call the native browser confirm() dialog. Buttons use onclick so the action is cancelled if the user clicks Cancel.

Planned JavaScript additions for Phase 3 include client-side form validation and dynamic date-range enforcement on the leave submission form.

6. API Integration Status

The application integrates with the Brevo Transactional Email API (v3) to send HTML email notifications whenever a leave request is approved or rejected.

Integration Point	Status	Notes
Brevo REST API call	Complete	POST to https://api.brevo.com/v3/smtp/email via PHP cURL
HTML email template	Complete	Branded CoreAxisHR template with leave details table
API key management	Complete	Key stored only in config.php, excluded from repo via .gitignore
Recipient resolution	Partial	Currently sends to MAIL_FROM; employee email column not yet added
Error handling on API fail	Pending	cURL errors are silently ignored — logging/fallback planned

Brevo was chosen for its free tier and straightforward JSON REST interface callable directly from PHP. A config.example.php placeholder is provided so collaborators can supply their own credentials without exposing secrets in version control.

7. Remaining Work Before Final Presentation

The following items are scoped for completion before the Phase 3 / final presentation:

Item	Description
Security hardening	Replace raw SQL strings with PDO prepared statements to eliminate SQL injection risk.
Password hashing	Switch from plain-text passwords to PHP password_hash() / password_verify().
Employee email column	Add an email field to the employees table so Brevo notifications reach the correct recipient.
Client-side validation	Add JavaScript form validation (required fields, date ordering) on leave submission.
Search on leave table	Extend the live search to the leave requests table for Admin/Manager views.
Dashboard stats	Add summary statistics to the dashboard (headcount, pending requests, etc.).
UI polish	Improve mobile responsiveness and refine empty-state messages across all tables.
API error handling	Log Brevo failures and show an in-app warning if the email could not be sent.

8. Brief Summary of Team Contributions

Oswaldo Cardona — Team Lead / Full-Stack & Back-End Developer Led overall project

Led overall project architecture and set up the XAMPP/PHP environment. Implemented the login/logout flow, session management, and the main dashboard. Coordinated task assignments and managed the GitHub repository, including the .gitignore and config.example.php credential strategy. Designed and created the MySQL database schema. Built the complete Employee CRUD module — add_employee.php, edit_employee.php, delete_employee.php, and employees.php. Wrote the leave request submission and retrieval logic in leave.php.

Samuel Mendoza — Front-End Developer & Back-End / API

Owned all front-end work: designed the CSS design system (style.css) including the sidebar layout, card components, badge styles, and color palette. Built the leave request submission form UI and ensured consistent CoreAxisHR branding across all pages. Integrated the Brevo Transactional Email API into leave_action.php, including writing the HTML email template and the cURL request logic. Managed config.php credential separation and tested the approve/reject workflow end-to-end.